

REINFORCEMENT LEARNING

SUBJECT CODE: 24DS742

Experiment 2: Interactive Exploration of Tic-Tac-Toe using Reinforcement Learning Lab ASSIGNMENT -2

Submitted by

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**MASTER OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**



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Task 1: Identify the Reinforcement Learning Components

Open the RL Tic-Tac-Toe demo and identify the following Reinforcement Learning components.

RL Component	Student Observation
Agent	A simple reinforcement learning algorithm for agent to learn the game tic-tac-toe.
Environment	A 3×3 tic-tac-toe board
State Representation	The current configuration of the board which of the 9 cells hold an X, an O, or are empty
Action Space	The set of empty cells available at a given turn; the agent chooses one to place its mark in
Reward Function	+1 for win, 0 for a draw, and -1 for losing
Learning Approach Used	Value-function-based reinforcement learning the agent estimates $V(s)$, the probability of eventually winning from states, and updates these estimates through repeated self-play

Who is the learning agent in this environment?

The learning agent in this environment is the player agent (AGENT X/O) which learn by training episodes, through simulation and through the feedback of moves it plays.

What is considered as the environment?

The 3x3 tic tac toe board game with empty spaces, rewards and the function values.

How is the game state represented?

The full 3×3 grid configuration, which is basically a snapshot of which cells will be filled with an X, O or nothing at that moment of the game.

What are the possible actions available to the agent?

Marking the empty spaces with O, getting feedback from the moves, learning from them and taking a win if there's a scope of placing 3 O's in a row or diagonal.

When does the agent receive a positive reward?

When it finishes 3 Os horizontally, vertically or diagonally (i.e. when it wins the game). States that stay closer to a win (such as two-in-a-row) don't get a direct reward, but rather high estimated value for them.

Which Reinforcement Learning approach is used for learning?

Value-function approach (temporal-difference style updates): The agent stores value functions and updates the value of states it regularly visits to gradually approach the realized outcome of its self-play games, instead of storing Q-values or deep neural networks.

Is the agent trained using labelled data?

No. There is no set of move, label pairs here. The agent will start playing the game with itself and will update its values based on what it played, whether it was a win, lose or draw, based on the rewards and not quite the same as supervised learning.

Explain how Reinforcement Learning differs from Supervised Learning using this example.

RL differs from supervised learning because it learns through interaction and rewards instead of labelled examples. In supervised learning you need a set of supervised examples with value and label pair. In here, the agent is not told what is labelled as 'correct move'. Rather it plays a variety of games, watches its results at the end of the game (win/loss/draw) and learns about which states and actions have good payoffs through trial and error and feedback on the rewards (win/loss/draw) instead of knowing them from the start.

Task 2 Observation Table

Episodes	Win %	Loss %	Draw %	Behaviour
100	65	5	30	Mostly random; easy to beat.
500	20	25	55	Blocks some threats; improving.
1000	40	20	40	More strategic and defensive.
5000	25	25	50	Strong play; many draws.
10000	15	50	35	Near-optimal; difficult to beat.

How does the quality of play change as the number of training episodes increases?

Quality of play improved as training increased. The win% has reduced significantly. Although the ratios between win, lose and draw are inconsistent, it can be said that the agent shifted from losing when it made mistakes towards playing the objectively correct, non-losing strategy which increasingly forces a draw against the opponent.

At approximately how many training episodes does the agent begin making intelligent decisions?

At around 500 - 5000 episodes as the ratio of loss to draw was seemingly high

At what stage does the learning appear to converge?

Learning converged around 5000-10000 episodes.

Why does the win percentage improve with additional training?

Between 500 and 5,000 episodes, the agent has run enough self-play games to build value estimates that reliably favor blocking threats and creating winning lines, so it starts capitalizing on any mistakes you make rather than playing randomly.

Why do most games end in a draw after sufficient training?

the agent's policy converges toward that optimal, non-losing strategy so instead of winning more, it increasingly neutralizes the game into a draw

What happens if the number of training episodes is too small?

Too little training causes random, poor decisions. learning isn't always smooth this early the value estimates are still noisy and haven't stabilized, so performance can temporarily dip before the agent has seen enough games to average out that noise

Conclusion

This experiment demonstrated how a Reinforcement Learning agent improves its performance through repeated self-play and experience. Initially, with fewer training episodes, the agent played randomly and made several mistakes, making it easier to defeat. As the number of training episodes increased, the agent learned to block opponent moves, select better positions, and make more strategic decisions. At higher training levels, the agent exhibited near-optimal gameplay, resulting in fewer losses and more drawn games. Overall, the experiment highlights that increasing training episodes significantly enhances the agent's learning, decision-making, and overall performance in the Tic-Tac-Toe environment.

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RL Demo Lab Assist

Reinforcement Learn

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jinglescode.github.io/reinforcement-learning-tic-tac-toe/

Reinforcement Learning Tic Tac Toe with Value Function

Demo

Train Agent

Number of Episodes

100

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 100)

Agent won. Reset game?

O	O	X
	O	
X	X	O

Agent's Value Function

0	0	0
50.0%	0	50.0%
0	0	100.0%

RESET GAME

Reinforcement Learning Tic

Notepad - Offline capable

jinglescode.github.io/reinforcement-learning-tic-tac-toe/

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Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 100)

Draw game. Reset game?

O	O	X
X	O	O
O	X	X

Agent's Value Function

0	0	0
0	0	0
25.0%	0	0

RESET GAME

Reinforcement Learning Tic Tac Toe with Value Function

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Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 100)

You won! Reset game?

O

O

X

O

O

X

X

X

Agent's Value Function

0

0

0

0

50.0%

50.0%

0

0

50.0%

RESET GAME

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Draw game. Reset game?

X

O

X

X

O

O

O

X

O

Agent's Value Function

0

0

0

0

0

0

0.0%

0

0

RESET GAME

Reinforcement Learning Tic

Notepad - Offline capable

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Reinforcement Learning Tic Tac Toe with Value Function

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You won! Reset game?

O

O

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X

X

X

Agent's Value Function

50.0%	0	50.0%
50.0%	0	50.0%
0	0	50.0%

RESET GAME